

Quantum Mechanics -- Introduction

Subject: Quantum Mechanics | Topic: Beginning of Quantum Mechanics

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1. Breakdown of Classical Radiation Physics

Towards the end of the nineteenth century, classical electromagnetic theory and equipartition of energy predicted an ultraviolet catastrophe for blackbody radiation. Rayleigh and Jeans treated cavity radiation as continuous standing waves, yielding an energy density proportional to frequency squared, which diverged as frequency grew.

2. Planck's Energy Quantization Hypothesis

In 1900, Max Planck proposed that microscopic atomic oscillators in cavity walls exchange energy exclusively in discrete packets (quanta): $E = n \cdot h \cdot \nu$, where n is an integer. This harmonic energy quantization smoothly suppressed high-frequency divergences and reproduced experimental blackbody spectra across all wavelengths.

3. Photoelectric Effect & Einstein's Photon

In 1905, Albert Einstein extended Planck's concept, proposing that light propagates and interacts as discrete energy packets: $E = h \cdot \nu$. The maximum kinetic energy of emitted photoelectrons satisfies: $K_{\text{max}} = h \cdot \nu - \Phi$, where Φ is the material work function.

4. Wave-Particle Duality & De Broglie Hypothesis

Louis de Broglie postulated that if radiant energy displays particle-like properties, material particles must symmetrically exhibit wave properties with wavelength: $\lambda = h / p$. This established the foundation for wave mechanics and the Schrodinger equation.